Handling Data 4 — Experimental and theoretical probability · Sample spaces

Give probabilities as fractions in their lowest terms unless asked otherwise. Marks in brackets. ★ = harder question.

1 Each face of a cube is painted red or blue: two faces red (R), four blue (B). The cube is rolled 12 times: B B R B B B R B B R B B. [7]

a Plot the relative frequency of red after 1, 2, 3, … 12 rolls on a graph.

b From the 12 rolls, estimate the probability of **(i)** a red face, **(ii)** a blue face.

c Comment on your results.

a Reds so far after each roll: 0,0,1,1,1,1,2,2,2,3,3,3 → relative frequencies $0, 0, 0.33, 0.25, 0.2, 0.17, 0.29, 0.25, 0.22, 0.3, 0.27, 0.25$. The graph jumps about at first and settles.

b (i) $\frac{3}{12} = \frac{1}{4}$ **(ii)** $\frac{9}{12} = \frac{3}{4}$.

c The theoretical probabilities are $\frac{2}{6} = \frac{1}{3}$ and $\frac{2}{3}$. The estimates are close but not equal — 12 rolls is a small sample; with many more rolls the relative frequency should settle near $\frac{1}{3}$.

2 A card is picked at random from an ordinary pack of 52. Find the probability that it is [5]

a red **b** a spade **c** not a queen **d** a black ace.

a $\frac{26}{52} = \frac{1}{2}$ **b** $\frac{13}{52} = \frac{1}{4}$ **c** $1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}$ **d** $\frac{2}{52} = \frac{1}{26}$.

3 A letter is chosen at random from the word ANTIDISESTABLISHMENTARIANISM (28 letters). Find the probability that it is [4]

a A **b** I **c** Z **d** a vowel.

Count: A×4, I×5, S×4, T×3, N×3, E×2, M×2, D, B, L, H, R (total 28). **a** $\frac{4}{28} = \frac{1}{7}$ **b** $\frac{5}{28}$ **c** 0 — impossible **d** A, I, E: $\frac{4+5+2}{28} = \frac{11}{28}$.

4 Two fair dice are thrown and the score is the *difference* between the two numbers (the larger minus the smaller). [9]

a Copy and complete the sample space showing all 36 outcomes.

–	1	2	3	4	5	6
1	0	1				
2		0		2		
3						
4						
5						
6						

b Find the probability that the score is **(i)** 0, **(ii)** at least 4, **(iii)** odd, **(iv)** a prime number.

c The game is played 72 times. How many odd scores would you expect?

a Each cell is $|\text{row} - \text{column}|$. Frequencies: score 0: 6 (the diagonal), 1: 10, 2: 8, 3: 6, 4: 4, 5: 2 — total 36.

b (i) $\frac{6}{36} = \frac{1}{6}$ **(ii)** scores 4 or 5: $\frac{6}{36} = \frac{1}{6}$ **(iii)** 1, 3, 5: $\frac{18}{36} = \frac{1}{2}$ **(iv)** 2, 3, 5: $\frac{16}{36} = \frac{4}{9}$.

c $\frac{1}{2} \times 72 = 36$.

Total: 25 marks

Check: theoretical probability needs *equally likely* outcomes · $P(\text{not } E) = 1 - P(E)$ · a sample space lists every outcome once — count carefully · expected frequency = probability × number of trials.